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1  # Note 19: PCA
2  # =====
3
4  # Import Libraries
5  import pandas as pd
6  from sklearn.preprocessing import StandardScaler
7  from sklearn.decomposition import PCA
8  import matplotlib.pyplot as plt
9
10 # Load dataset
11 df = pd.read_csv("data.csv")
12
13 # Select features
14 X = df[["Feature1", "Feature2", "Feature3", "Feature4"]]
15
16 # Standardize
17 scaler = StandardScaler()
18 X_scaled = scaler.fit_transform(X)
19
20 # Apply PCA
21 pca = PCA(n_components=2)
22 X_pca = pca.fit_transform(X_scaled)
23
24 print("PCA Data:")
25 print(X_pca)
26
27 print("Explained Variance Ratio:")
28 print(pca.explained_variance_ratio_)
29 print("PCA Components : ", pca.components_)
30
31 print("Total Variance:")
32 print(pca.explained_variance_ratio_.sum())
33
34 # Find Number of Components
35 pca = PCA(n_components=0.95)
36 X_pca = pca.fit_transform(X_scaled)
37 print("Number of components:",
38       pca.n_components_)
39 print("Explained variance:",
40       pca.explained_variance_ratio_.sum())
41
42 # PCA Visualization
43 plt.scatter(X_pca[:, 0], X_pca[:, 1])
44 plt.xlabel("PC1")
45 plt.ylabel("PC2")
46 plt.title("PCA")
47 plt.show()
48
49
```